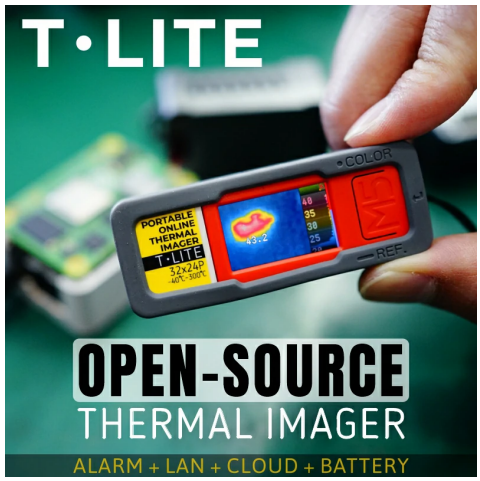
T-Lite

SKU:K126





















Description

The `T-Lite` is an integrated `thermal imaging` temperature measurement device with online detection capabilities. This product is primarily composed of the `M5StickC-Plus` and `HAT THERMAL`, utilizing the `MLX90640-BAA` infrared imaging sensor. The main control is powered by the `ESP32-PICO-D4` chip, featuring `WIFI` functionality, a `135 x 240` resolution `TFT` screen, and a built-in `160mAh battery`. The name "T-Lite" represents Temperature (T) and Lite, signifying compactness and high integration. Users can set up a `high-temperature warning function` and view temperature images and data in real-time via `cloud` and `local network`. M5 also provides an online data API interface, allowing users to access remote images and corresponding data through `EZData`, facilitating engineering project applications.

Features

- Developed based on ESP32-PICO, WiFi
- User button, LCD (1.14 inch), power/reset button
- 160 mAh lithium battery
- Integrated passive buzzer
- Field of view: 110°×75°
- Temperature measurement range: -40°C ~ 300°C

Includes

- 1 × T-Lite
- 1 × StickC-Plus
- 1 × Lanyard
- 1 × USB Type-C cable (50cm)
- 1 × User manual

Applications

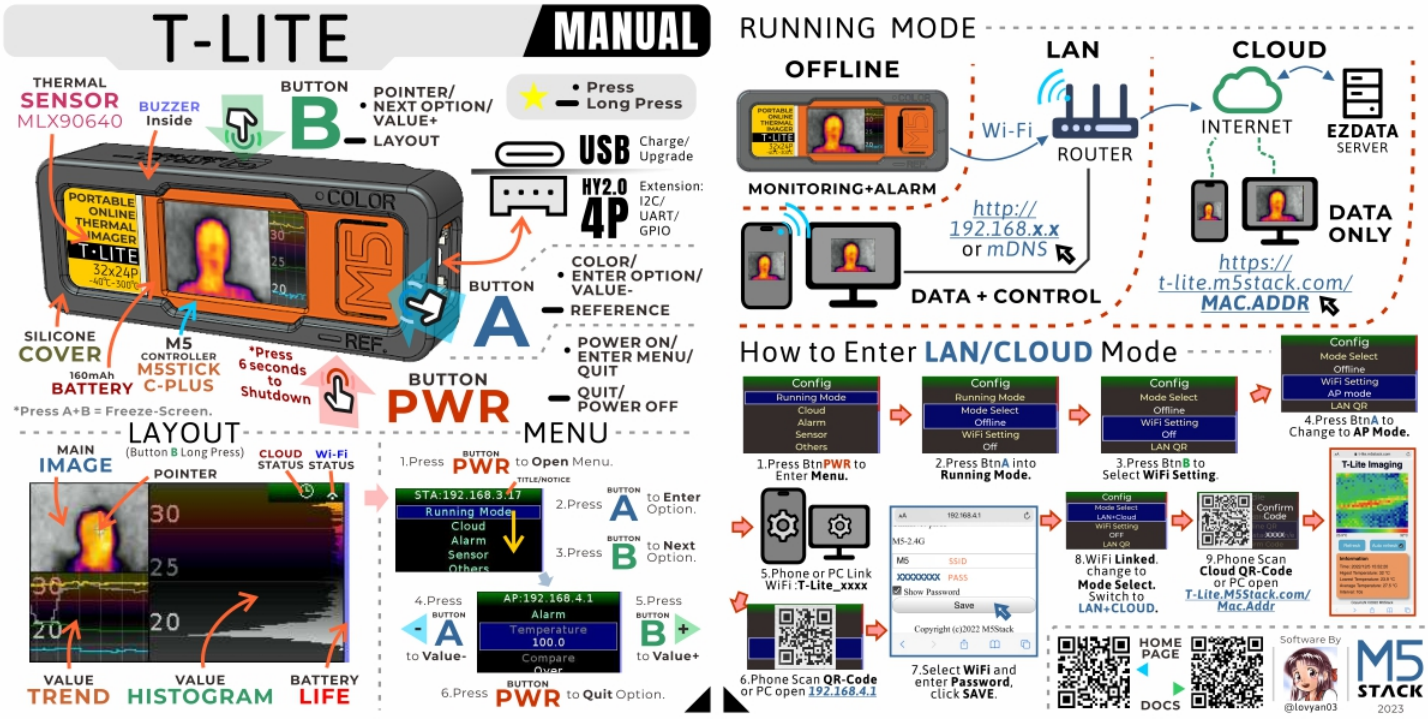
- Temperature detection and warning
- Circuit temperature anomaly monitoring
- High-precision non-contact temperature measurement
- Biological movement monitoring
- Visual infrared imaging

Specifications

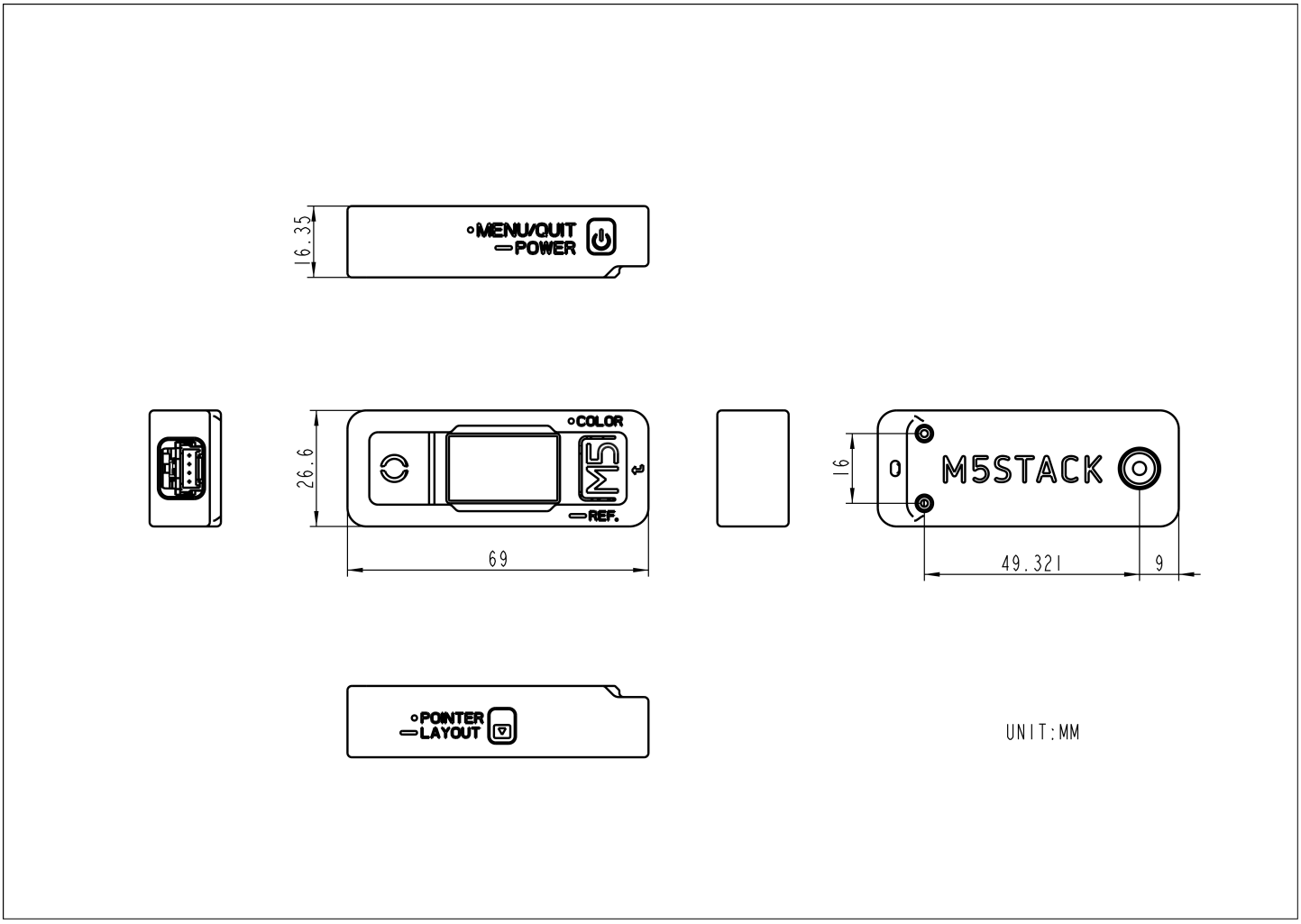
Resources	Parameters
MCU	ESP32-PICO,240MHz dual core, 600 DMIPS, 520KB SRAM
Sensor	MLX90640
Field of View	110°×75°
Resolution	32×24
Flash Memory	4MB Flash
Power Input	5V @ 500mA
Port	Type-C
LCD screen	1.14 inch, 135 x 240 Colorful TFT LCD, ST7789v2
MIC	SPM1423
Buzzer	Passive buzzer
PMU	AXP192
Battery	160 mAh @ 3.7V
Product Size	69.0 x 26.6 x 16.4mm
Product Weight	31.0g
Package Size	110.0 x 87.0 x 25.0mm
Gross Weight	62.9g

User Manual

Electronic Manual



Model Size



Datasheets

- MLX90640

Softwares

Quick Start

- T-LiTe User Manual

PlatformIO

- T-Lite Firmware

EasyLoader

Easyloader	Download Link	Notes
T-Lite Firmware Easyloader	download	/

Emissivity Modification Guide

Emissivity (ε)

Infrared sensors default to an ideal blackbody (ε=1.0), but most materials reflect ambient radiation, causing measurement errors. By setting the correct emissivity (ρ = 1–ε), you can compensate for reflections and improve measurement accuracy.

Material	Emissivity ε
Human skin	0.98
Water	0.93
Plastic (opaque)	0.95
Asphalt	0.95
Concrete	0.95
Brick	0.90
Glass (plate)	0.85
Wood (natural)	0.94
Aluminum (oxidized)	0.30
Steel (oxidized)	0.80

- For more emissivity/reflectivity data, see: Emissivity Reference Table (PDF)

Note: Be sure to set the appropriate `emissivity` parameter in your API or library calls based on the target material to ensure measurement accuracy. In the source code src/mlx90640.cpp, the default emissivity is hard-coded in the `calcTempData()` function:

```
1141 void MLX90640_Class::calcTempData(const uint16_t *framedata,
1142                                    temp_data_t *tempdata,
1143                                    const temp_data_t *prev_tempdata,
1144                                    uint32_t filter_level, uint8_t monitor_width,
1145                                    uint8_t monitor_height) {
1146     float emissivity = 0.95;  // ← Original line: 1146 — change to desired ε value

1148     float Ta = MLX90640_params.MLX90640_GetTa(framedata);
1149     float tr = Ta - TA_SHIFT;
1150     bool subpage = framedata[833];
1151     tempdata->subpage = subpage;

1153     MLX90640_params.MLX90640_CalculateTo(
1154       framedata, emissivity, tr,
1155       tempdata, prev_tempdata, filter_level);

1157     // …additional processing…
1158 }
```

After modifying line 1146 (`float emissivity = 0.95;`) to your target emissivity, save and recompile to apply the change.

Video

- Product Tutorial

T-Lite(en).mp4